Memory OS of AI Agent

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PART 1

The Problem

Part 1 **The Problem · long-term memory in multi-session conversation**

Part 2 Existing Memory Methods and Their Failure Modes

Part 3 MemoryOS · the idea and the full procedure

Part 4 Experiments and Takeaways

Long-Term Memory in Multi-Session Conversation



- **fact recall** · answer from what was said weeks ago, across sessions
- **personalization** · act on what the user wants without being told again
- **one memory per user** · spans separate sessions · topics grouped by the system, not by the user

Question Types in Long-Term Conversation Benchmarks

LoCoMo

≈ 300 turns
≈ 9K tokens / conversation



questions asked after the whole conversation
scored by F1 and BLEU-1 against a gold answer

GVD 15 virtual users × 10 days · at least two topics per day · an LLM judge scores retrieval accuracy, correctness, coherence

single-hop

p1 park | p2 squirrels | p3 run

one page

Where did the user go last week?

needs · one page

answer · **the wetland park**

multi-hop

p1 park | p2 squirrels | p3 run

two pages, linked

What did the user see at the place where they went running?

needs · two pages joined

answer · **squirrels**

temporal

p1 park | p2 squirrels | p3 run

T · week 1 + today's date

How many weeks ago did the user visit the park?

needs · a page plus its timestamp

answer · **three weeks ago**

open-domain

p1 park | p2 squirrels | p3 run

+ world knowledge · + preferences

What would be a good gift for the user?

needs · pages plus outside knowledge

answer · **running shoes**

Fixed Context Window as the Root Cause

week 1 · park, squirrels, ran two laps … weeks of other conversations, ≈ 9K tokens and growing … **today · "remember the park?"**

the whole conversation, in tokens



outside the window → gone
"Sorry, I don't remember you mentioning the park."

a bigger window is not a fix
conversations keep growing · and nothing says who the user is

so a memory system has to answer four questions

1 what to store	2 how to update	3 what to recall	4 how to answer
becomes the storage module	becomes the updating module	becomes the retrieval module	becomes the generation module
tiers, pages, topic folders	what moves where, what is dropped	which pages return for a query	how recalled memory enters the prompt

– **root cause** · the model sees only what fits in the window, and earlier sessions do not

PART 2

Existing Memory Methods

Part 1 The Problem · long-term memory in multi-session conversation

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Three Families of Memory Methods

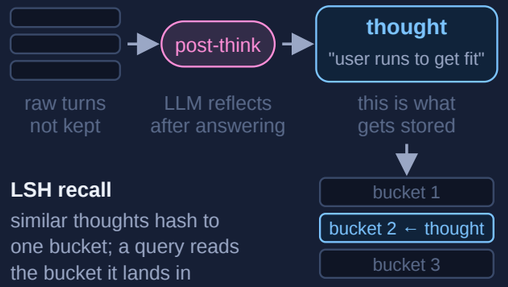
knowledge-organization
structure what the model has thought or noted

retrieval-mechanism-oriented
an external library and a recall rule

architecture-driven
change the control flow that reads and writes

TiM

Think-in-Memory · Liu et al. 2023



LSH recall
similar thoughts hash to one bucket; a query reads the bucket it lands in

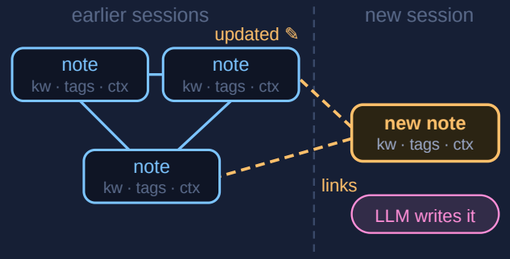
memory ops on thoughts · insert · merge · forget

stores reasoning outcomes, not raw turns
recall by LSH over stored thoughts
memory ops · insert, forget, merge

focus · **storage structure**

A-Mem

Agentic Memory · Xu et al. 2025



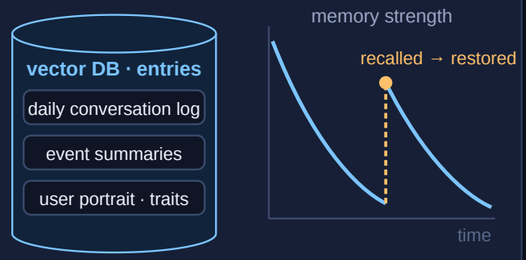
memory evolution
a new note is linked to related notes and may rewrite their context and tags

LLM writes a structured note per turn
links it into a knowledge network
old notes evolve when new ones arrive

focus · **storage structure**

MemoryBank

Zhong et al. 2024



each entry · embedding + time
Ebbinghaus forgetting curve

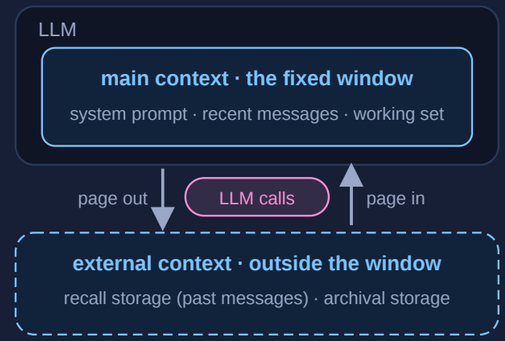
recall · query embedding vs entries; weak entries fade out, recalled ones are refreshed

logs conversations, events, user traits
strength decays on an Ebbinghaus curve
refreshed on recall · portrait grows

focus · **retrieval mechanism**

MemGPT

Packer et al. 2023



the model itself decides when to read and write

two contexts in an OS-like hierarchy
explicit read / write calls page data
past the fixed window

focus · **control flow**

– **four baselines** in the experiments · each family focuses on one dimension of memory management

Failure Modes of Existing Memory Methods

1 MemGPT · flat FIFO queue

pages of every topic share one queue,
so topics mix as the dialogue grows

in →



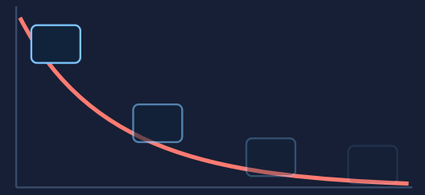
one queue · topics interleaved

oldest out, regardless of topic

OS-style paging, but no notion of a topic

2 MemoryBank · decay alone

a forgetting curve decides what fades,
but nothing organizes or updates the rest



strength only decays · no structure, no update

worst scores of all compared methods

3 TiM · single-stage hash retrieval

one hash lookup returns similar thoughts,
but cannot preserve cross-topic dependencies



the link between topics is lost

one lookup, one bucket at a time

saving thoughts beats raw turns, retrieval is flat

4 A-Mem · heavy link generation

every note needs multi-step LLM calls to link,
which inflates latency and accumulates errors

LLM call × 13 per response



each step can inject an error

rich graph, slow and fragile to build

13.0 LLM calls per response (Table 3)

(MemoryOS: 4.9 calls)

each method works on a single dimension in isolation · storage structure, retrieval mechanism, or update strategy

no unified system that stores, updates, retrieves and generates together

PART 3

MemoryOS

Part 1 The Problem · long-term memory in multi-session conversation

Part 2 Existing Memory Methods and Their Failure Modes

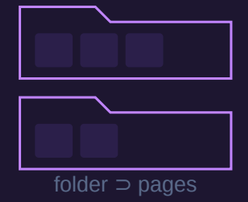
Part 3 MemoryOS · the idea and the full procedure

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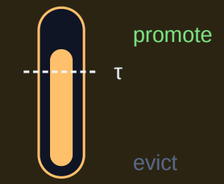
Operating System Principles Reused in MemoryOS

operating system	MemoryOS
segment–page memory logical segments subdivided into fixed pages (Multics) segment metadata → no external fragmentation paging → little internal fragmentation	**topic folders of dialogue pages** a page = one turn {Q, R, T} · a folder = one topic folder carries an LLM summary and keywords pages join the folder whose match score is highest
priority-based eviction keeps hot data LRU, working-set models: recently and often used data stays resident, the rest is paged out coarse segments + fine pages minimize overhead	**heat-based eviction and promotion** heat = visits + number of pages + recency coldest folder is evicted when the store is full hot folder (heat > τ) is promoted to the persona tier
storage hierarchy cache → main memory → disk small and fast near the processor, large and slow further away	**three memory tiers by lifetime** **STM** recent turns → **MTM** topic folders → **LPM** persona short-term · mid-term · long-term personal what moves between tiers is decided by FIFO and heat







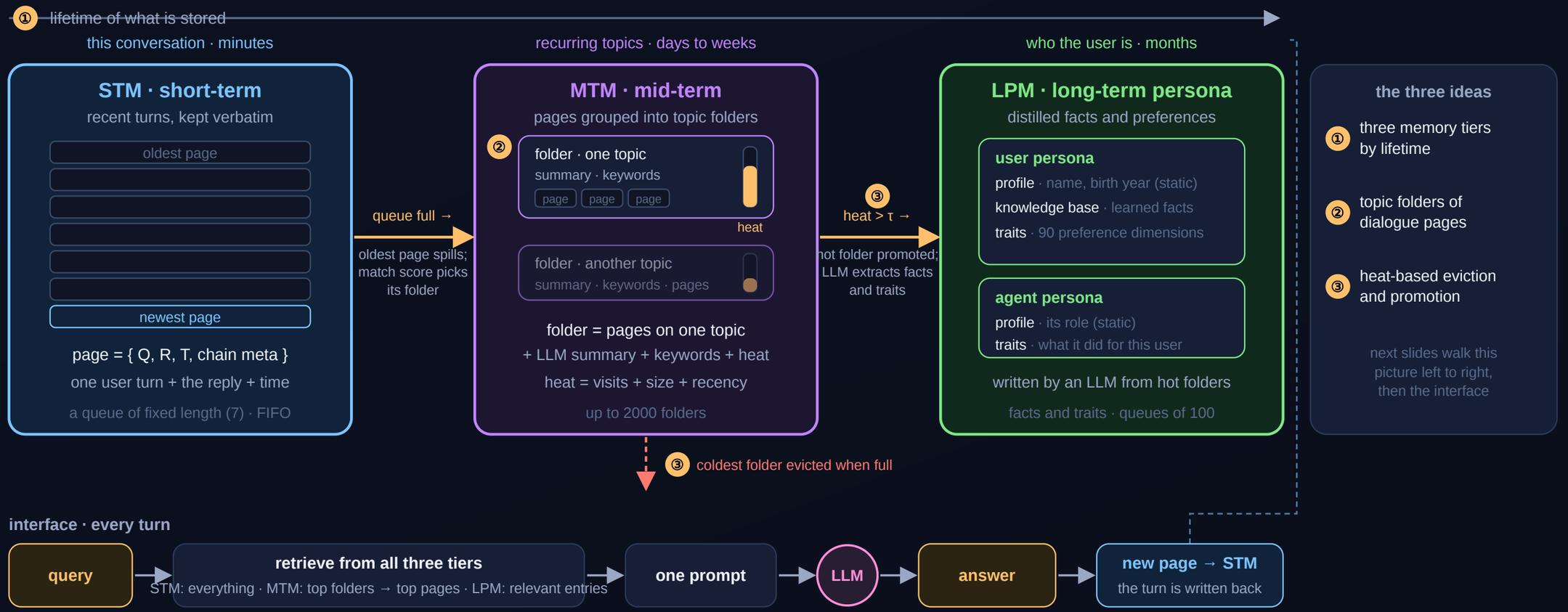






– **same three principles** · segments of pages · keep the hot, evict the cold · tiers by lifetime

Structure of MemoryOS



everything above runs between the user's message and the reply; the LLM is called for the answer and for the summaries and extractions inside the tiers

Running Example Used Throughout Part 3

week 1 · session 1
one visit to a park

p1 I went to the wetland park last week, the scenery was beautiful.

p2 Many little squirrels were running in the trees, very cute.

p3 I ran two laps in the park. I'll keep working out to get slimmer.

→ topic folder · wetland park outing

three consecutive turns on one thread

…

week 2 · session 2
a separate chat

p4 Any good movies to watch this weekend?

p5 Is there a cinema near my home?

→ topic folder · weekend movie plans

…

week 3 · session 3
three unrelated topics

p6 A quick pasta recipe for dinner tonight?

→ topic folder · weekend cooking

p7 I'll go to the gym three times a week from now on.

→ topic folder · gym schedule

p8 I'm flying to Busan next month.

→ topic folder · flight to Busan

…

week 4 · today
the turn we will trace

p9 Remember the wetland park I told you about?

p10 I feel like eating a burger, any recommendations?

a good answer to p10
"The cheeseburger at […], just don't forget you're hitting the gym!"

needs p3 and p7 · three and one weeks old, from other sessions

how to read the next slides

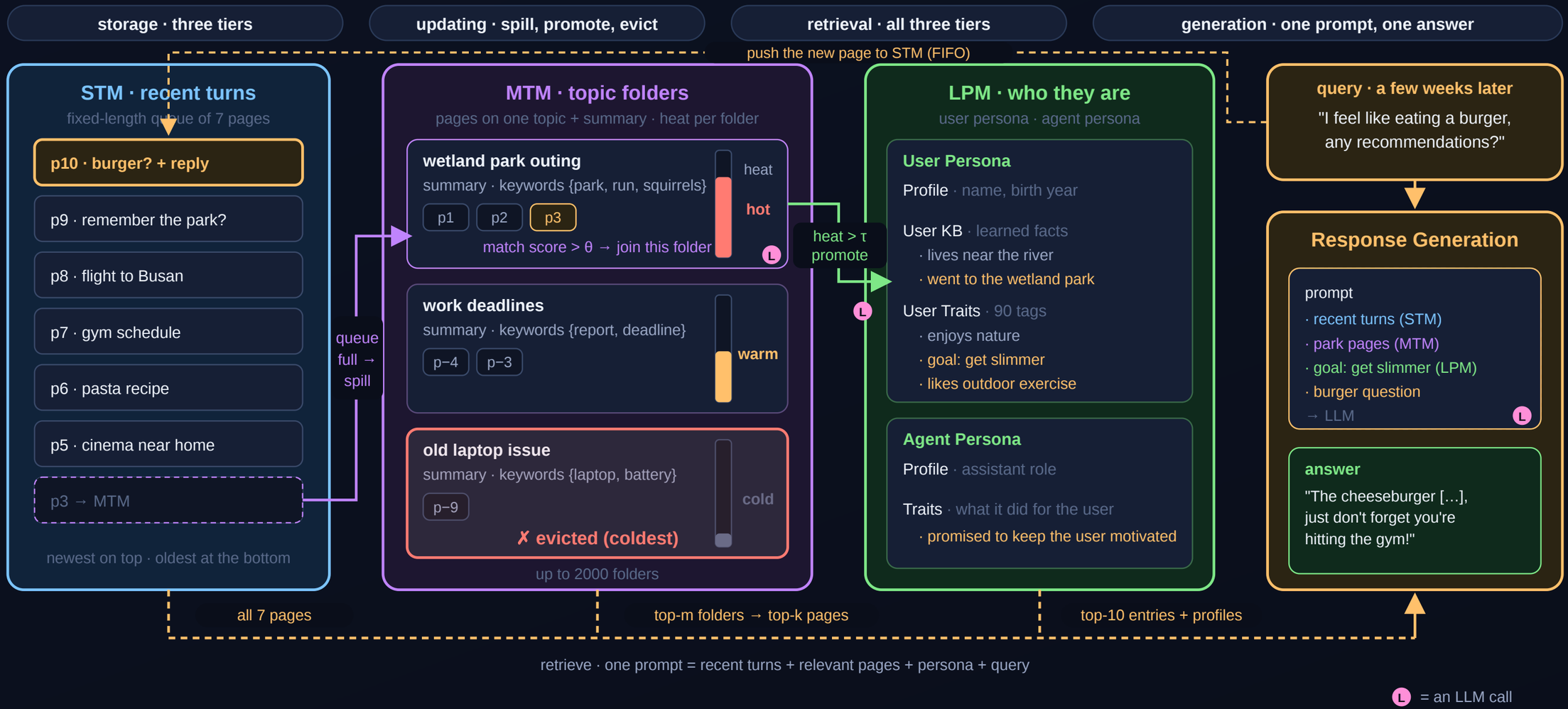
p1 … p10 are the same pages on every slide · **purple** = the topic folder a page ends up in · **amber** = today's query and what moves because of it

older topics from before week 1 are already in memory (work deadlines, favourite restaurants, recipe ideas, an old laptop issue); their pages are numbered backwards, p−1, p−2, …

the recent queue holds 7 pages, so p1, p2, p3 leave it one by one as p8, p9, p10 arrive, and join their topic folder

– **one example, start to finish** · a four-week conversation, traced through every part of the structure

MemoryOS on the Running Example



Three Storage Tiers and Their Contents



snapshot · week 1, right after p3

STM · recent turns

page = {Q, R, T} · pages on one thread form a chain

p1 T · week 1, Mon 19:02
Q went to the wetland park, scenery beautiful
R sounds like a relaxing experience

p2 T · week 1, Mon 19:04
Q many little squirrels in the trees, very cute
R squirrels signal a healthy environment

p3 T · week 1, Mon 19:07
Q ran two laps, will keep working out to get slimmer
R exercise is good, I'll keep you motivated

chain

chain meta · running summary, written by an LLM (L)
"user visited a wetland park, saw squirrels,
ran two laps, wants to get slimmer"

snapshot · week 4, after p1–p3 left the queue

MTM · topic folders

folder (segment) = pages on one topic + summary + keywords

wetland park outing — heat · hot

summary (L)
"a park visit with squirrels and a run;
the user wants to get in shape"

keywords
park · squirrels · run · slimmer

pages
- p1 park
- p2 squirrels
- p3 run

work deadlines · 2 pages · warm

old laptop issue · 1 page · cold

snapshot · week 4, after the folder was promoted

LPM · who the user and the agent are

fixed fields stay · learned fields grow (FIFO queues of 100)

User Persona

Profile static — name · gender · birth year

User KB dynamic — facts learned from the chat
- went to the wetland park (week 1)
- runs laps in the park

User Traits dynamic — 90 tags in 3 groups
- goal: get slimmer
- likes outdoor exercise

needs & personality · AI alignment · content interests

Agent Persona

Profile static — role: a friendly health-minded assistant

Traits dynamic — what it did for this user
- promised to keep the user motivated

– **page** {Q, R, T} → **topic folder** + summary + keywords → **persona** facts and traits

Dialogue Pages and the Dialogue Chain



LLM ① link or reset **LLM ② summarize the chain**

page = { Q, R, T, meta }

a few weeks later

p1 T · week 1, 18:02
Q I went to the wetland park last week, the scenery was beautiful.
R Sounds like a relaxing experience!
meta → chain summary below

p2 T · week 1, 18:04
Q There were many little squirrels running in the trees, very cute.
R I like squirrels, they signal a healthy environment.

p3 T · week 1, 18:07
Q I ran two laps and felt great. I'll keep working out to get slimmer.
R Exercise is good for health. I'll keep you motivated!

p10 T · week 4, 12:31
Q I feel like eating a burger, any recommendations?
R The cheeseburger [...], just don't forget you're hitting the gym!

① continues p1? ✓ link **① continues p2? ✓ link** **① continues p3? ✗ reset**

chain 1 · meta = summary of all pages on the chain, rewritten by LLM ② at every link

wetland park visit last week, beautiful scenery
+ saw many squirrels in the trees (a healthy environment)
+ ran two laps there, felt great · goal: keep working out to get slimmer

chain 2 · new chain starts at p10

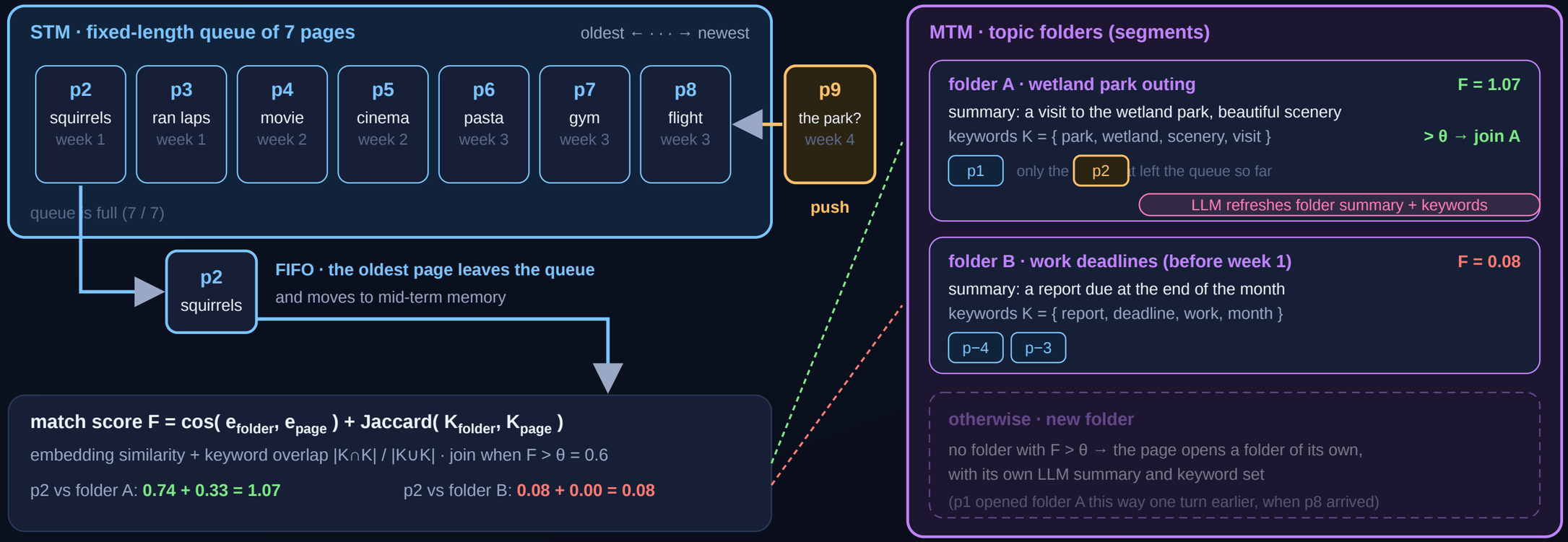
wants a burger recommendation
(meta of chain 1 is kept with p1–p3)

STM holds pages in arrival order; the chain adds "which earlier pages this one continues" and a summary, so a page never loses its thread

– **chain** · pages on one thread, linked or reset by the LLM · running summary kept as **meta**

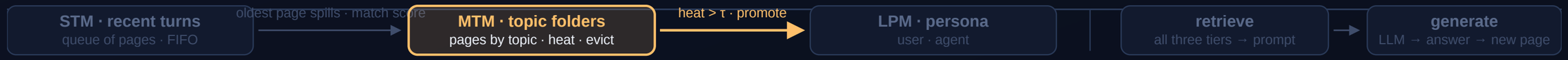
Update From Short-Term to Mid-Term Memory





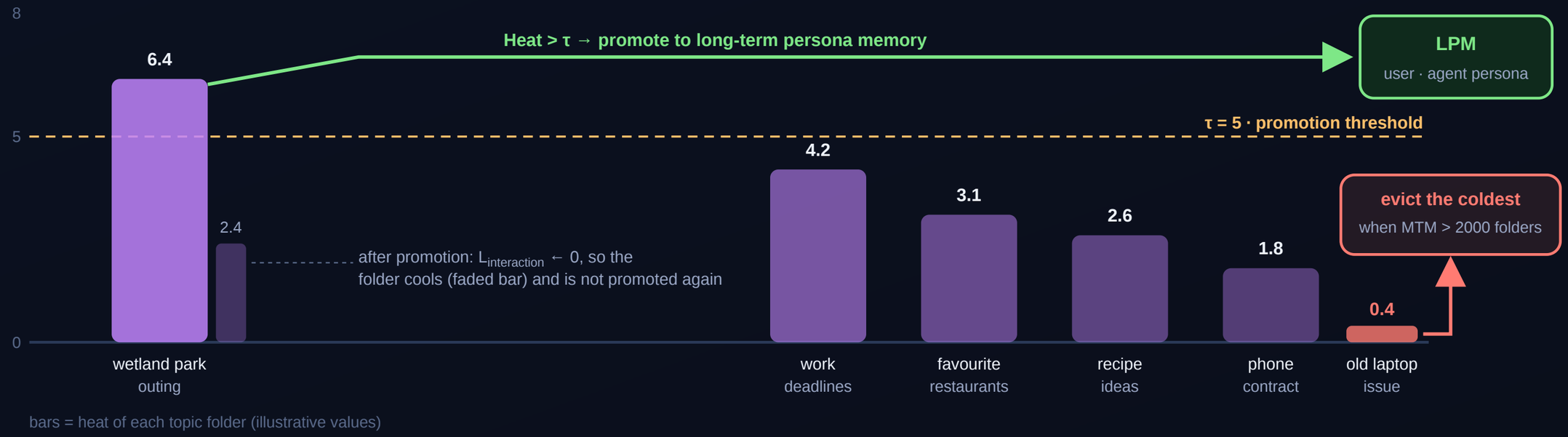
– **FIFO** out of the 7-page queue · **match score** F > θ joins a topic folder, otherwise a new folder

Segment Heat, Eviction, and Promotion



$$\text{Heat} = \alpha \cdot N_{\text{visit}} + \beta \cdot L_{\text{interaction}} + \gamma \cdot R_{\text{recency}}$$

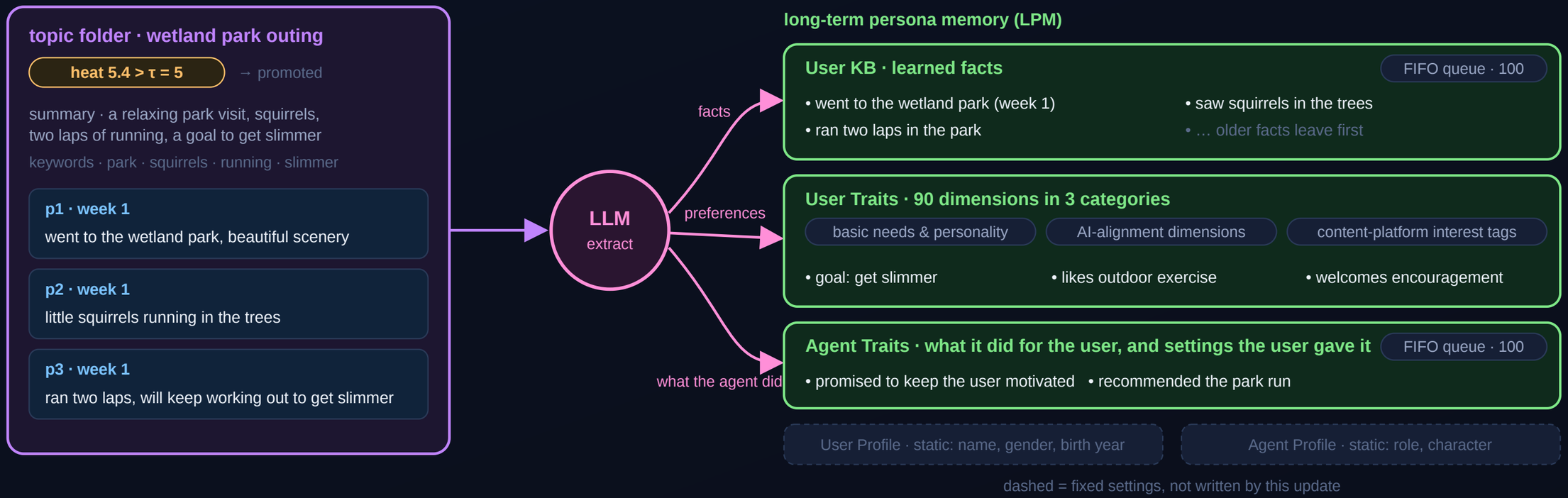
$\alpha = \beta = \gamma = 1 \cdot \mu = 10^7$ s — times the folder was retrieved — number of pages in the folder — $\exp(-\Delta t / \mu)$, Δt since last access



- **promote** · Heat > τ = 5 → the folder's pages and summary feed the long-term persona memory
- **evict** · when MTM exceeds 2000 folders, the coldest one is deleted

Update to Long-Term Persona Memory

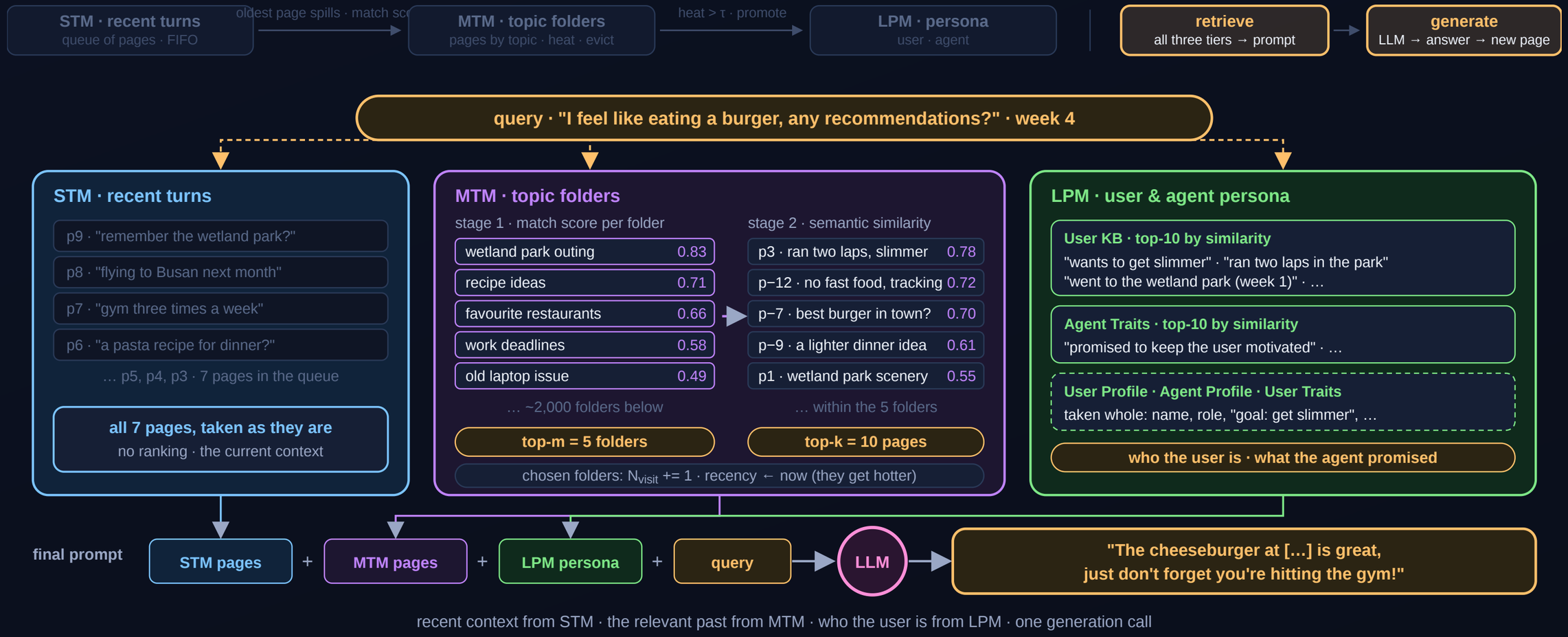




after the transfer the folder's $L_{interaction}$ resets to 0, so its heat drops and the same content is not promoted twice

- **heat > τ** · one LLM pass writes facts, 90-dim traits, and agent traits · queues of 100

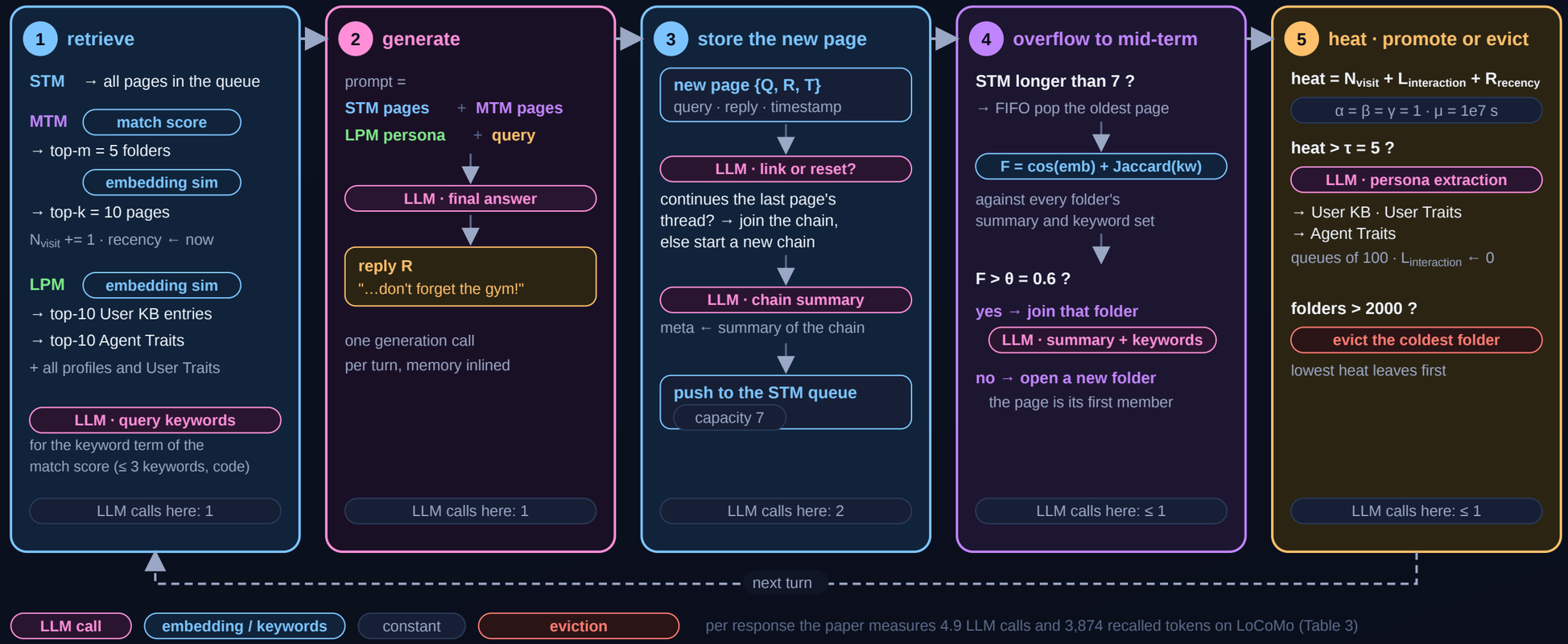
Memory Retrieval and Response Generation



recent context from STM · the relevant past from MTM · who the user is from LPM · one generation call

The Per-Turn Procedure of MemoryOS





- **LLM calls per turn** · link judge · summaries · persona extraction · the answer

PART 4

Experiments

Part 1 The Problem · long-term memory in multi-session conversation

Part 2 Existing Memory Methods and Their Failure Modes

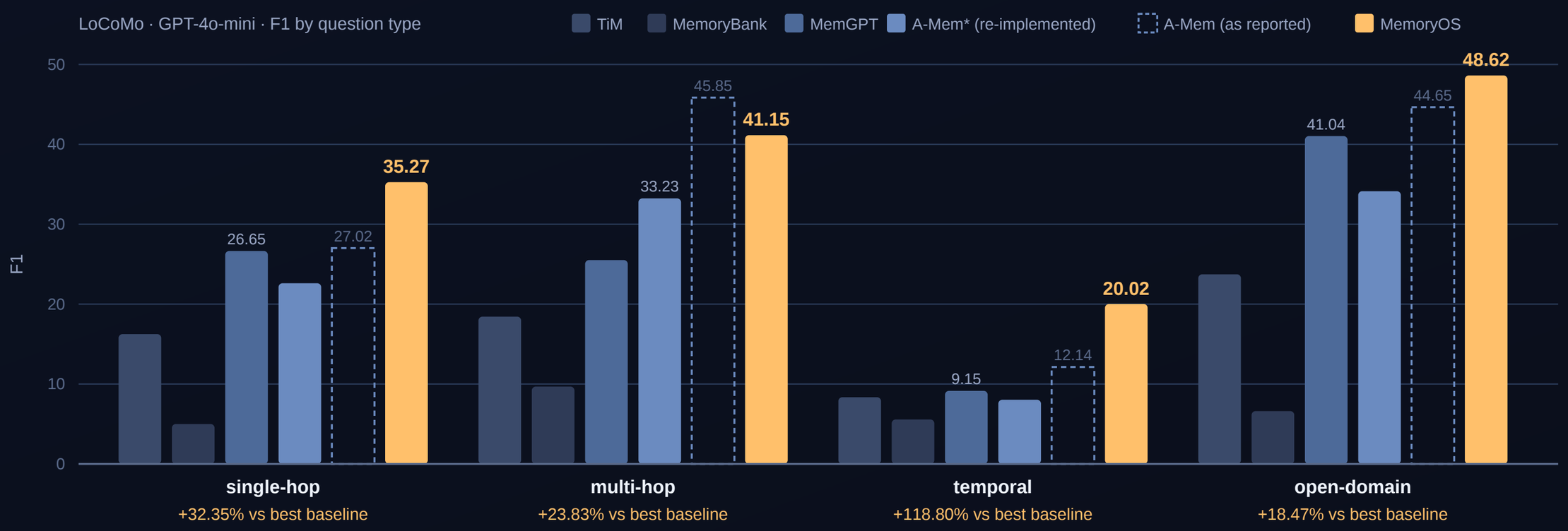
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Experimental Setup

	GVD	LoCoMo
conversations	15 virtual users × 10 days · ≥ 2 topics per day · simulated user–assistant dialogues	≈ 300 turns · ≈ 9K tokens per conversation · questions in 4 types: single-hop · multi-hop · temporal · open-domain
metrics	retrieval accuracy (0 / 1) · correctness and coherence (0 / 0.5 / 1) · scored by DeepSeek-R1	F1 and BLEU-1 against the reference answer
backbones	GPT-4o-mini · Qwen2.5-7B	GPT-4o-mini · Qwen2.5-3B
baselines	TiM · MemoryBank · MemGPT · A-Mem (A-Mem* = re-implementation in the same environment; A-Mem = numbers reported by its authors)	
MemoryOS settings	recent turns queue 7 pages · topic folders ≤ 2000 · User KB / Agent Traits queues 100 · promotion $\tau = 5$ · folder match $\theta = 0.6$ · heat weights $\alpha = \beta = \gamma = 1$ · $\mu = 1e7$ s retrieval: top-m = 5 folders · top-k = 5 (GVD) / 10 (LoCoMo) pages · top-10 persona entries	
hardware	8 × H20 GPUs	

Results on the LoCoMo Benchmark



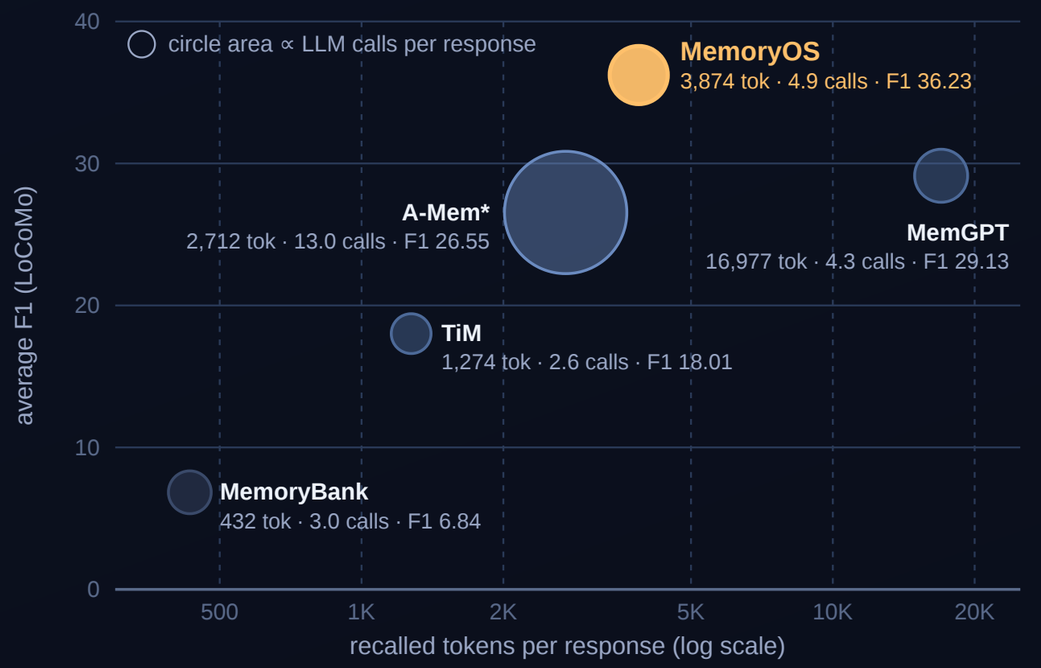
average over the four types vs the best baseline: +49.11% F1 · +46.18% BLEU-1 · average rank 1.0 on both metrics

* on multi-hop, the A-Mem value its authors report (45.85) exceeds MemoryOS (41.15); the re-implementation under the same environment gives 33.23, which is the basis of the +23.83%

Results on GVD and Efficiency

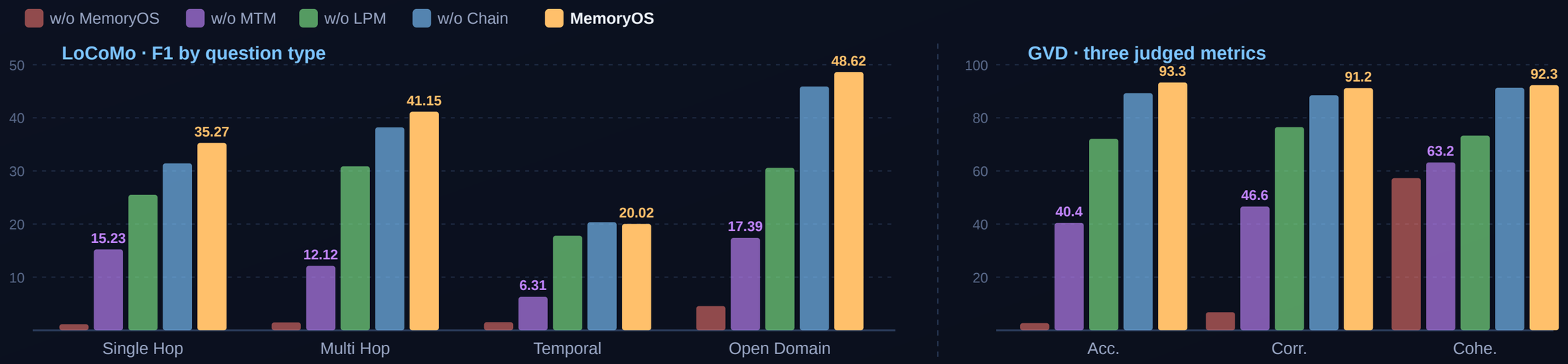
backbone	method	Acc.	Corr.	Cohe.
GPT-4o-mini	TiM	84.5	78.8	90.8
	MemoryBank	78.4	73.3	91.2
	MemGPT	87.9	83.2	89.6
	A-Mem	90.4	86.5	91.4
	MemoryOS	**93.3**	**91.2**	**92.3**
Qwen2.5-7B	TiM	82.2	73.2	85.5
	MemoryBank	76.3	70.3	82.7
	MemGPT	85.1	80.2	86.9
	A-Mem	87.2	79.5	87.8
	MemoryOS	**91.8**	**82.3**	**90.5**

GVD · retrieval accuracy / correctness / coherence, scored by DeepSeek-R1 · MemoryOS over A-Mem with GPT-4o-mini: +3.2% / +5.4% / +1.0%



fewer calls than A-Mem* (4.9 vs 13.0) · far fewer tokens than MemGPT (3,874 vs 16,977)

Ablation over the Memory Modules





k = 10 adopted · gains flatten beyond 20

single-hop F1 on LoCoMo · 5 → 10 is the big step, 20 → 40 is flat or down (noise from extra pages)

footnote · on temporal questions the chain is the one exception: 20.34 without it vs 20.02 with it · all values GPT-4o-mini, Fig. 2 and Fig. 3

– **MTM** matters most · **LPM** next · **chain** least · every tier removed costs something on every question type

Takeaways

Hierarchy, topic folders, heat

recent turns → topic folders → persona
FIFO moves a page down
match score picks its folder
heat promotes hot folders, evicts cold ones
four modules in one per-turn loop

Where the gains come from

LoCoMo avg F1 **+49.11%** over baselines
temporal, the hardest type: **+118.8%** F1
ablation: **MTM** > LPM > chain
the win is organization, not a bigger prompt

Cost and limits

4.9 LLM calls · **3.9K** tokens per reply
A-Mem* 13 calls · MemGPT 17K tokens
hand-set sizes: STM 7 · 2000 folders · τ = 5
overlapping folders never merged
GVD is LLM-judged · A-Mem re-run lower

Discussion · **heat** rewards what was used often; when a user's interests shift, which signal should decide what to forget, and does a fixed **90-dimension trait schema** carry over to other domains?